

# Motion Detector

ApplicationsPlaces

Terminal

editViewSearchTerminalHelp

```
type P 1429
ed
28 93 37 10 41 5 6 34 6 0
er Motion detection events for frame 43 frame type P
ON START
type P 1263
ed
29 113 15 22 58 8 2 17 0 2
er Motion detection events for frame 44 frame type P
ON START
type P 1229
ed
17 96 23 15 37 5 6 41 4 5
er Motion detection events for frame 45 frame type P
type P 1449
ed
27 110 29 14 70 15 5 48 4 3
er Motion detection events for frame 46 frame type P
ON COMPLETE
type P 1875
ed
48 98 20 31 69 14 10 50 7 5
er Motion detection events for frame 47 frame type P
ON COMPLETE
```

black&white

contour

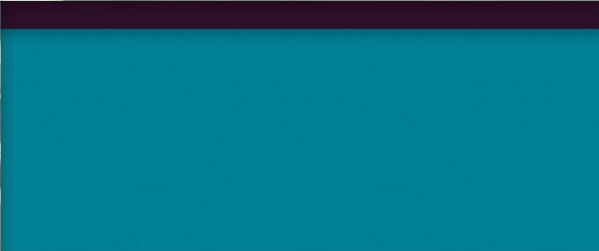
motion vectors

occupancy

Double Right Click inside polygon to delete Polygon







[(51,947 unre...]

[New Tab - Go...]

Terminal

[/experiment/...]

[/experiment...]

[untitled(adh...]

Terminal

occupancy

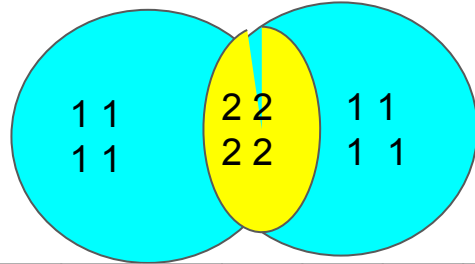
black&white

contour

motion vectors

# MD

1. Average occupancy for last 3 frames
2. Open CV::findcontour
3. Vector Direction



|  |  |   |   |   |   |   |  |
|--|--|---|---|---|---|---|--|
|  |  |   |   |   |   |   |  |
|  |  |   |   |   |   |   |  |
|  |  |   | 1 | 1 | 1 |   |  |
|  |  | 1 | 1 | 1 | 1 | 1 |  |
|  |  | 1 | 1 | 1 | 1 | 1 |  |
|  |  |   | 1 | 1 | 1 |   |  |
|  |  |   |   |   |   |   |  |
|  |  |   |   |   |   |   |  |

|   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|--|
|   |   |   |   |   |   |   |  |
|   |   |   |   |   |   |   |  |
|   |   | 1 | 1 | 1 | 1 | 1 |  |
| 1 | 1 | 2 | 2 | 1 | 1 | 1 |  |
| 1 | 1 | 2 | 2 | 1 | 1 | 1 |  |
|   |   | 1 | 1 | 1 | 1 | 1 |  |
|   |   |   |   |   |   |   |  |

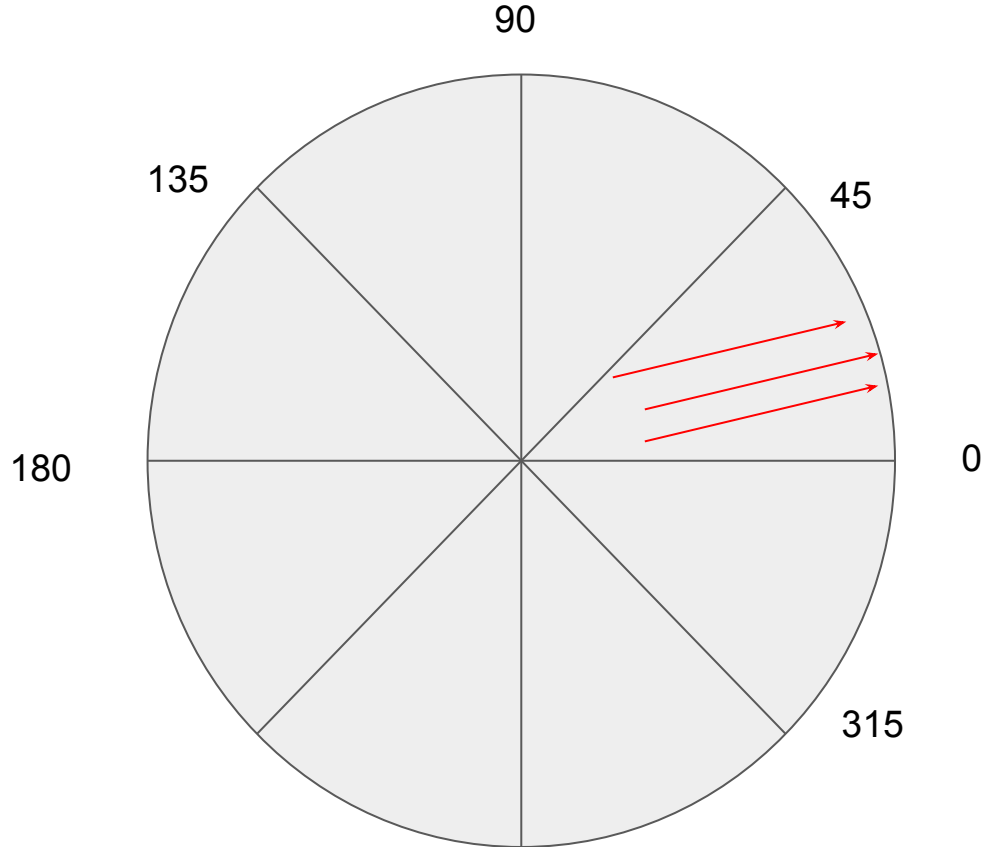
|   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|--|
|   |   |   |   |   |   |   |  |
|   |   |   |   |   |   |   |  |
|   | 1 | 1 | 1 | 0 | 0 |   |  |
| 1 | 1 | 1 | 1 | 1 | 0 | 0 |  |
| 1 | 1 | 1 | 1 | 1 | 0 | 0 |  |
|   | 1 | 1 | 1 | 0 | 0 |   |  |
|   |   |   |   |   |   |   |  |
|   |   |   |   |   |   |   |  |

# Open CV::findcontour

Tracking of moving ball with  
Binary CV::MAT

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 |
| 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

# Vector Direction



# Add, Edit Camera and Masking coordinates

Onfly, change the masking coordinates with Rest API

```
"Potential ID": "729795",
"rtcMaxPort": 12560,
"Rtc MinPort": 11501,
"pre_clip": 5,
"rtsp": {
  "0": {
    "rtsp": "rtsp://root:60056005@x.x.x:554/axis-media/media.amp?videocodec=h264&resolution=1280x720&fps=25",
    "state": "streaming",
    "CLIPSIZE": 20,
    "DEFAULT_WINDOW_SIZE": 3,
    "DEFAULT_OCCUPANCY_THRESHOLD": 2,
    "DEFAULT_LOCAL_OCCUPANCY_AVG_THRESHOLD": 0.6,
    "DEFAULT_OCCUPANCY_AVG_THRESHOLD": 0.6,
    "md": true,
    "masking_coords" : [{"x" : 0,"y" : 0}, {"x" :1280,"y" : 0}, {"x" : 1280, "y" : 720}, {"x" : 0,"y" :720}]
  },
}
```

# Security

- Auth Token Authorization
- HTTPS
- Secured Rest APIs

# Segments Having motions

I P P P P I P P P P I P P P P I —> FPS 5 - 25  
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 -  
M M M

Motion happened at 8 and 9 frames ( Triggers MD at 8 frames

Seg 1- 5 will not be send

Seg 6- 10 will be only send and saved ( FMp2) +- delta ( pre 2 Secs , post 18 Secs) after trigger at 9th frame for coming 20 Sec  
Motion Extraction will stop.

Seg 11- 15 will not be send

9 th frame trigger



# Preclip config before motions

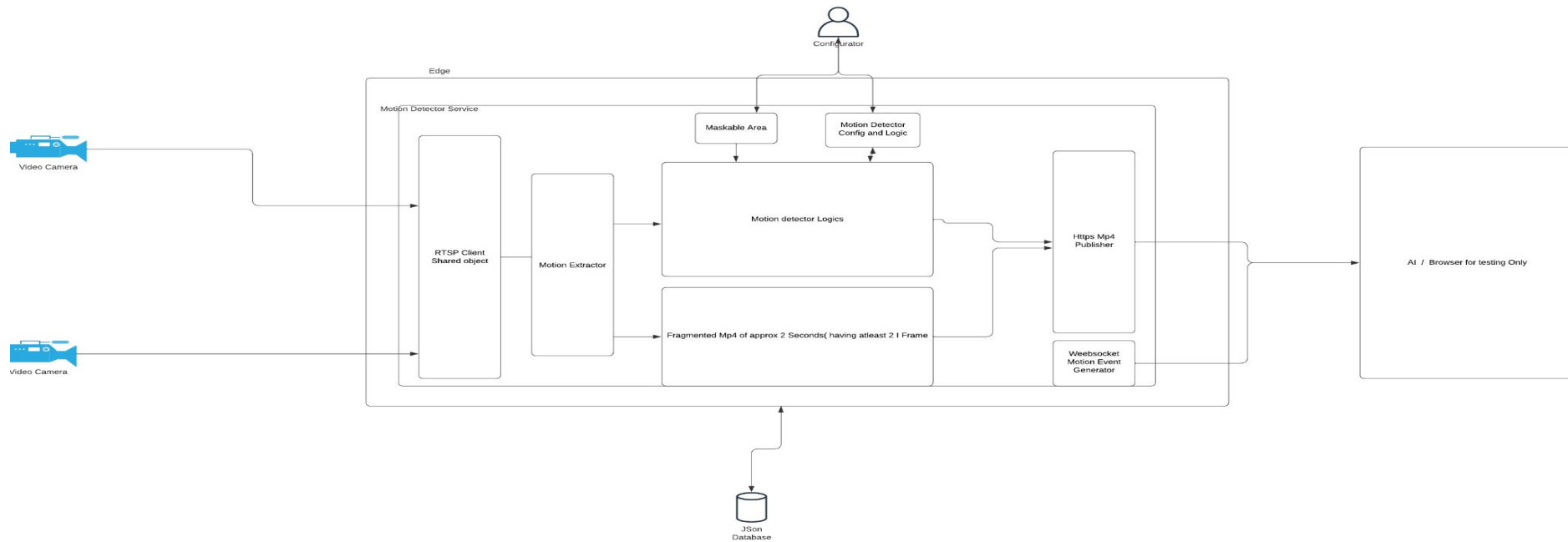
- When GOP is 5 and FPS is 5, If Pre-clip configured as 5 in the code then MP4 will start from 21 frame which is I frame.
- Below mentioned clips elaboration for the reference

1 6 11 16 21 MMM

- \* When GOP is 10 and FPS is 5, If Pre-clip configured as 5 in the code then MP4 will start from 31 frame which is I frame.
- Below mentioned clips elaboration for the reference

[illegible]

**When GOP size is 10 or 33 etc PreClip size can never be accurate. It will be approx around 4-6 seconds**



### Info

*Motion Detector Services can take RTSP streams of 250 camera.*

*Motion Detector Service will have components Motion Extractor, Motion Detector Logic, Configurator, Fragmented Mp4 Publisher and Motion Events*

*Configurator can change the Motion Detection logic and Maskable Area*

*JSON DB for storing Motion Event, time, clip and camera detail. It will be done in Third phase.*



# Motion Detection Service

Readme.md

<https://bitbucket.org/webrtc-solution/mediaserver/src/provigil/src/motiondetection/>

Service Name: mdserver

## Modules

1) RTSP Client Shared , Get RTSP Stream from Camera ( it will be same as existing Mpeg Dash and Webrtc Service) Camera will be added by Rest API as we do in Mpeg Dash and Webrtc Service

2) FPS? FPS will be the same as Camera FPS.

3) Why 2 Seconds? it will not be exactly 2 seconds. It will have around 2 I frames and 48 P or B frames. If Camera has FPS 25 , then 2 Seconds Clips will have around 50 Frames with 2 I frames and 48 P or B frames Also it will be configurable from 2 Second to 10 seconds Fragmented mp4. Fragmented MP4 can be combined when duration of motion clips are longer during publishing. Cost of buffering and storing longer duration Mp4 will be high, so we are going for a smaller duration Mp4.

# Configuration

FPS - RTSP FPS 5 - 25 will be best

Respective Area ( this area will keep changing with resolution) some factor value

Number of P frame changing 2-3 best

Threshold ( Signama of  $MV[1-Area]$  > .05 ) P frame has motion

MV AI

MV



# MV

Motion Vector will be Vectors of [ X1, Y1, X2, Y2 D] ( start and end point of coordinates ) Where D is Forward ( P or B Frame) or Backward prediction( B Fame)

Motion Detection ( MD) Algorithms. It will be a collection of MD algorithms. Which will return true or false. Will be provided by Configurator( Chandra) as a Shared Object plugin. User can Add new Maskable area and Select MD algorithms with Rest APIs

```
bool MD1( Vectors of [ X1, Y1, X2, Y2 D] , Polygon of Masked Area )
```

```
{
```

```
    summation of Vectors of [ X1, Y1, X2, Y2 D]) of masked area
```

Extracted motion vector is passed to process\_frame() in MV Detector. Inside the function, the occupancy map is determined based on the motion vectors. The occupancy\_map is 8x8 grid and deemed to be occupied when the motion vector originating the grid cell is found.

```
}
```

```
bool MD2( Vectors of [ X1, Y1, X2, Y2 D] , Polygon of Masked Area, Time T1 )
```

```
{
```

# We are planning to give a demo of Motion Detection in 2 phases.

## **Phase1 Demo .**

- a) Motion Vector Extraction
- b) Motion Detection Events.
- c) Fragmented FMP4 storage and streaming

## **Phase2 Demo.**

- a) Rest API for Masking Area
- b) Configuration of Motion Detection Algorithm
- c) Motion Detection form Motion Vectors , Advance Algorithms ( Chandra and team will provide advance algorithm)
- d) MP4 publishing

**Phase3 could be for patents and Provigil Algorithms**

# Motion Event

## Triggers

Camera\_ID\_2022\_02\_22:04:04:02\_3444

After an occupancy map for each frame is found, the average occupancy for last N frames is calculated. Which aims to reduce temporarily noise. The default value is set to 3 (MVDetector::DEFAULT\_WINDOW\_SIZE).

At each frame the total number of grid in the occupancy map whose values is above a threshold, currently defaulted to 0.6 (MVDetector::DEFAULT\_LOCAL\_OCCUPANCY\_AVG\_THRESHOLD), is counted. When this number is above a threshold (DEFAULT\_OCCUPANCY\_THRESHOLD), the frame is finally decided to contain a movement`.



# What is required for testing

Need one or two cameras where motion happen at fixed interval

Cam1 with FPS 5 motion happen every one mins

Cam2 with FPS 25 motion happen every 2 mins

## Results

MP4 will have only one second clip for one min

# Platform MDServer provides

OpenCV will be provide for backward compatibility

MP4 clips

JSON DB

Rest API for configuration of MD, mask area and selection of MD algorithm

On the fly change of MD algorithm